

Chance machine

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming
Lesson	Chance machine (lesson 6 of the unit)
Lesson ID	pf-106
Experience ID	pf-106
Plan ID	PLAN-PF-L06
Content version	d2745940dcb5
Website	https://revise360.co.uk/experience.html?id=pf-l06

Big question, outcomes and success criteria

Big question — How does a program produce a number that nobody can predict?

Prerequisites

The lesson before this one is pf-105. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pf-101, pf-102, pf-103, pf-104.

Assessable outcomes

1. **pf-l06-o1** — know what random numbers are used for in a program
2. **pf-l06-o2** — understand how a random whole number is generated within a given range
3. **pf-l06-o3** — know that a computer produces pseudo-random numbers from an algorithm and a seed

Vocabulary, resources and preparation

Key vocabulary — Pseudo-random, Seed

Resources

SLIDES	presentations/PF_L06_ChanceMachine_Lesson.pptx
WORKSHEET	worksheets/PF_L06_ChanceMachine_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pf-106

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.

2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on PF_L06_ChanceMachine_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Why random numbers.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Why random numbers

- A random number lets a program behave differently each time it runs, so the result cannot be predicted.
- Games use them for dice rolls, card shuffles and where an enemy appears.
- Simulations, random sampling, shuffling a list and generating a password all need them too.

Challenge on this wall: Give two reasons a dice game would be no fun if the program produced the same number every time.

2. Generating a random number

- In the OCR Exam Reference Language, `random(1, 6)` returns a random whole number from 1 to 6, and both end points are included.
- The value returned is normally stored in a variable, so the program can use that one number more than once.
- Calling the function a second time gives a fresh number, which may or may not be the same as the first.

Challenge on this wall: Write two lines in the OCR Exam Reference Language that roll a dice and output the result.

3. Choosing the range

- The two values passed to the function set the lowest and the highest number that can be returned.
- In Python 3, `random.randint(1, 6)` can return 6, but `random.randrange(1, 6)` stops at 5.
- An array of five elements is indexed 0 to 4, so a random index must come from that range and no wider.

Challenge on this wall: An array holds eight prize names. State the range a random index must come from, and say what would happen if the range went up to 8.

4. Pseudo-random

- A computer cannot be truly random: it works each number out with an algorithm, so the numbers are called pseudo-random.
- The algorithm starts from a value called the seed, and the same seed always produces the same sequence of numbers.
- Most languages take the seed from the computer's clock, so the sequence is different each time the program runs.

Challenge on this wall: Explain why a programmer testing a game might choose to set the seed to the same value every time.

5. Random in a loop

- A new number is generated every time the function is called, so a loop can produce a whole set of them.
- In the OCR Exam Reference Language a count-controlled loop suits a known number of values, such as ten dice rolls.
- Each call is independent of the last, so the same value can come up twice in a row.

Challenge on this wall: Describe what a program would need to do to count how many sixes it rolls in one hundred dice rolls.

6. Using randomness well

- A simulation uses random numbers to model something unpredictable, such as traffic or the weather, so it can be run many times.
- Random sampling picks members of a group at random, so the sample is fair and not chosen on purpose.
- A shuffle puts a list into a random order, and a password generator picks random characters.

Challenge on this wall: Suggest how a program could pick three pupils at random from a class list without picking the same pupil twice.

Retrieval and the check for understanding

Starter — A quiz program picks one of twenty questions to ask first. Write down what the program must do to choose a different question each time it runs, and the lowest and highest number it needs.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf-106-s4-c1), so the wording a pupil sees matches the wording you say.

Why are the numbers a computer produces described as pseudo-random?

Correct: They are worked out by an algorithm, so they are not truly unpredictable

What a pupil might choose instead

- They are always the same number
- They can only ever be between 0 and 1

- They are chosen by the user rather than by the program

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: An algorithm follows fixed steps, so it can only imitate randomness, however convincing the imitation looks.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Guided	14	Support is in the activity: an example to run, an output to predict, one thing to change, a gap to fill.
Independent	0	The pupil writes or fixes it with the brief and nothing else.
Challenge	0	The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Every activity in this lesson is guided: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Why random numbers	2	Random numbers are used when...
2	Generating a random number	2	In the OCR Exam Reference Language, random(1, 6) returns...
3	Choosing the range	2	The end points of the range matter because...
4	Pseudo-random	2	Pseudo-random means...
5	Random in a loop	2	Calling a random function inside a loop...
6	Using randomness well	2	A simulation uses random numbers to...
final	Final challenge: how many rolls	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pf-106-x1	State two different reasons why a program might need a random number.	2	worksheet, Exam practice a); deck, Exam practice slide
pf-106-x2	An array called prize holds eight prizes. Describe how a random number could be used to choose one of them, and state the range that number must come from.	3	worksheet, Exam practice b); deck, Exam practice slide
pf-106-x3	Explain why the numbers produced by a computer are described as pseudo-random.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in answers/records/pf-106.json, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Why does a game use random numbers?** — So the game runs faster; So the game uses less memory
- **Which of these jobs needs a random number?** — Working out the average of six marks; Counting how many questions a pupil answered
- **In the OCR Exam Reference Language, which values can random(1, 6) return?** — Any whole number from 1 to 5; Any whole number from 2 to 5
- **Why is the result of random(1, 6) usually stored in a variable?** — So that the user cannot change the number; Because random(1, 6) cannot be written on its own
- **In Python 3, which numbers can random.randrange(1, 6) return?** — 1, 2, 3, 4, 5 or 6; 2, 3, 4, 5 or 6

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- What random numbers are used for in a program
- How to generate a random number in a given range, and why the end points matter
- What pseudo-random means, and what a seed does

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pf-107, Revision HQ 2.2. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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